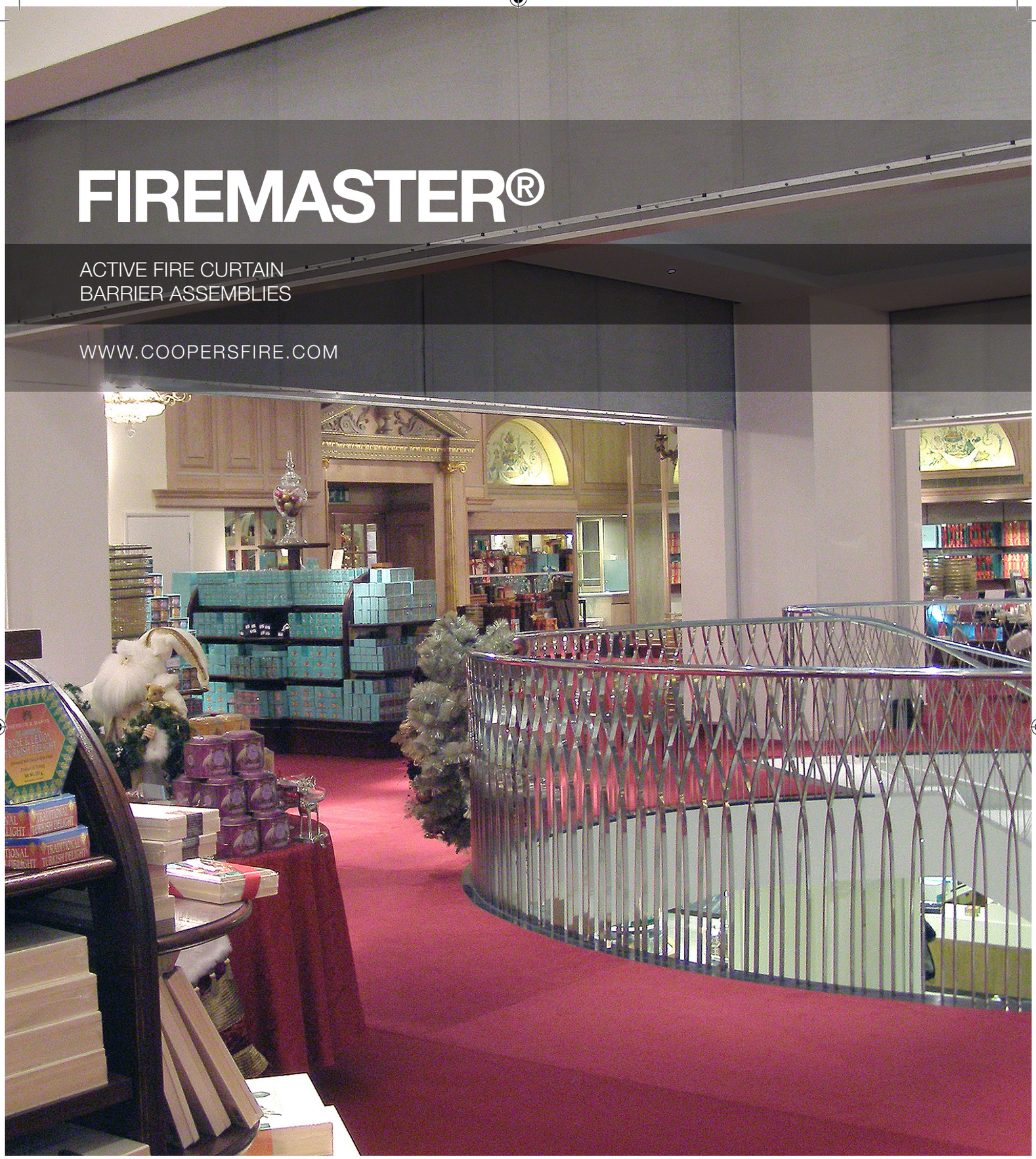
FIREMASTER®

ACTIVE FIRE CURTAIN
BARRIER ASSEMBLIES

WWW.COOPERSFIRE.COM



coopers
EST. 1983

LEADING THE WAY IN FIRE PROTECTION

DESIGN SOLUTIONS
THAT WORK FOR YOU

Active Fire Curtain Barrier Assemblies comprise technologically advanced fire-resistant fabric barriers encased in a compact steel housing. Barriers remain invisibly retracted until activated by an alarm or detector signal, at which time they descend safely to their fire operational position.

To be compliant, Fire Barriers must be protected from a short circuit and or total power failure, this is achieved with our patented Total Gravity Fail-Safe (TGFS) system.



Fire barriers will stop and control the spread of fire in a building. If means of escape protection is required use FireMaster® Plus².



PRODUCT ADVANTAGES

- Single roller assembly curtains for spans up to 5m as standard. For single roller assemblies up to 17m wide, use our FireMaster® Wide Single Roller (WSR)
- All barriers allow egress for occupants and access for emergency services
- Total Gravity Fail-Safe (TGFS) operation ensures safe rates of descent even following total power loss, system corruption or wiring short or open circuit
- Unique overlapping design with continuous, conjoined bottom bar for unlimited widths







SHOP FRONTS
- No need for expensive and noticeable fire-rated glass

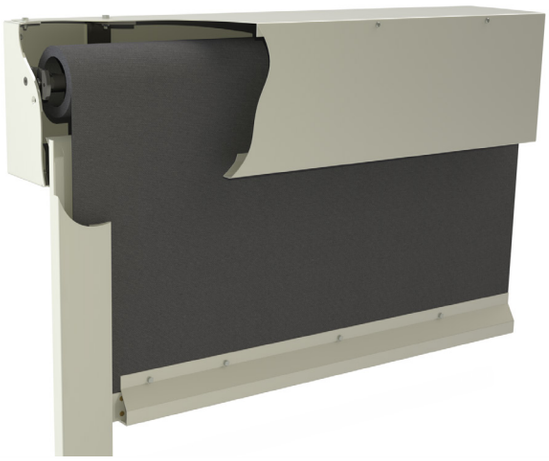
ATRIA
- Curtains allow multi-floor concourse openings for large atria

ESCALATORS & STAIRWELLS
- No need for fire-rated walls

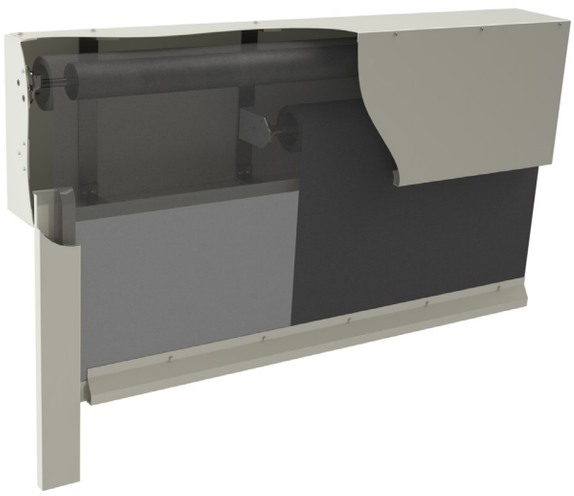
COMPARTMENTATION
- Virtual Fire Walls with unlimited width for maximum visibility to create open space

BOUNDARY PROTECTION
- Eliminate vertical and horizontal fire spread to other buildings

MEANS OF ESCAPE
- If means of escape are required use FireMaster® Plus[2]



Single Roller Assembly



Multiple Roller Assembly

- Multiple roller assembly curtains for spans over 5m as standard
- Fully synchronised intelligent drive system with ‘slave line’ for multiple rollers
- Unlimited spans with drops of up to 8m

CASE STUDY: FORTNUM & MASON

LONDON UK

As part of its tercentenary celebrations, Fortum & Mason undertook a £24 million refurbishment that involved the removal of inhibiting walls and doors to open up the stores retail space.

To remain compliant with fire regulations, it was necessary to ensure that the building could continue to contain a fire, but meeting fire regulations whilst realising the open plan design would call for non-evasive fire protection.

Coopers solution was to install 16 FireMaster® active fire curtain barriers at the entrance of each trading floor that would remain hidden within the ceiling until required.

On activation, all 16 fire curtains deploy to containing the spread of fire for up to 2 hours. By installing Coopers Fire curtains, Fortum & Mason were able to safely remove the stores non-loadbearing walls and fire doors and ensure that the new, open plan atrium complied with fire regulations

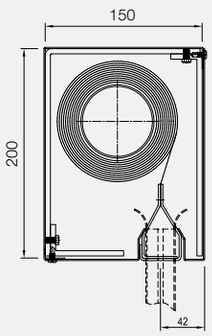
BENEFITS

- 16 FireMaster® fire curtains installed to contain a fire for up to 2 hours.
- Patented SLAT™ ceiling system ensures a sympathetic alignment
- Ensured open plan atrium met fire regulations
- Improved customer migration
- Fire Curtains branded with corporate logos

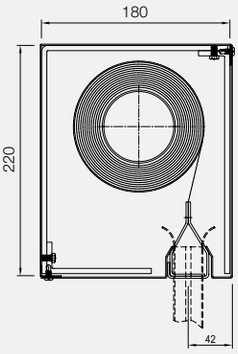
STANDARD HEADBOX SIZES

Single rollers

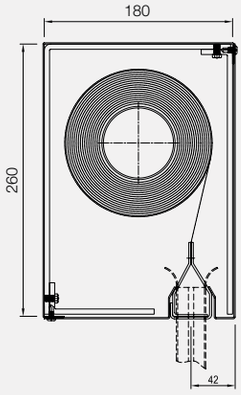
FM 15/20



FM 18/22

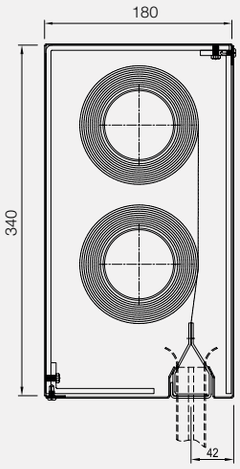


FM 18/26

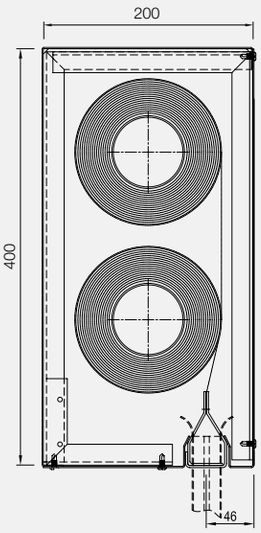


Multiple rollers

FM 18/34



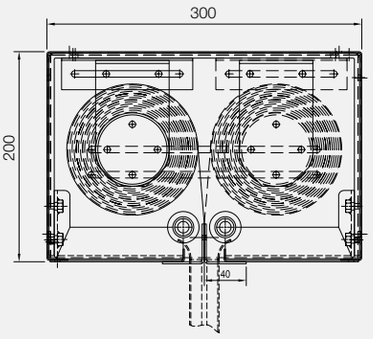
FM 20/40



Model No.	Maximum Barrier Width up to	Maximum Barrier Drop up to
FM 15/20	5 metres	4 metres
FM 18/22	5 metres	6 metres
FM 18/26	5 metres	8 metres
FM 18/34	Unlimited	4 metres
FM 20/40	Unlimited	8 metres

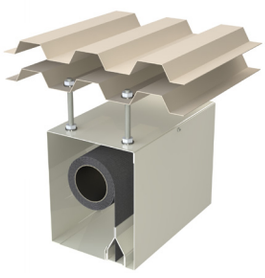
Coopers has a design solution for any size requirement. See the chart to the left for our standard range of sizes. For queries about special size requirements, please contact our technical department.

FM 30/20
Side By Side

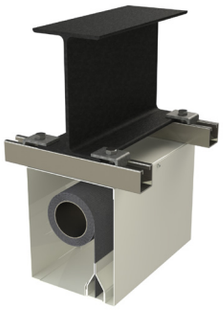


HOW THE HEADBOXES ARE INSTALLED

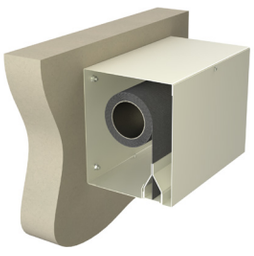
Unistrut and Drop Rods



I Beam



Face Fixed



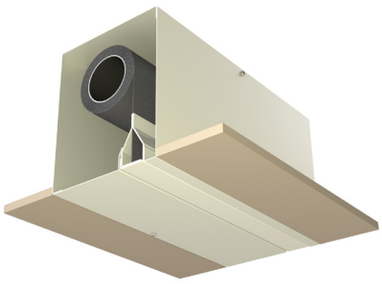
Soffit Fixed

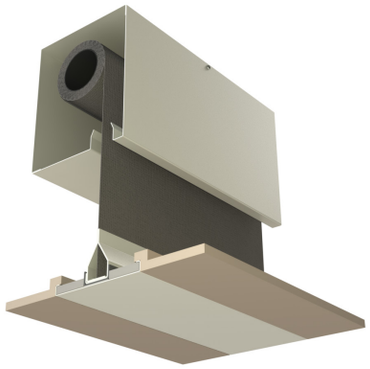


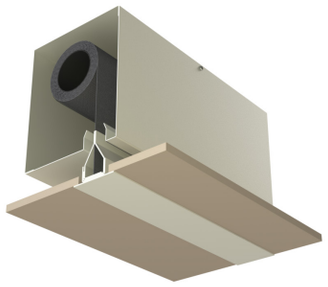
There are many fixing options to suit all types of ceiling configurations. Barriers can be integrated with both solid and suspended ceilings, enabling total project design flexibility.

HOW ARE THE BARRIERS CONCEALED?

FireMaster® systems provide a range of ceiling interfaces and can even invisibly conceal the barriers in the ceiling whilst still allowing access for service and maintenance. All our systems remain totally concealed when non-operational.







For illustrative purposes only

Flush Fit	Suspended ceiling with SLAT™ (Self Levelling Access Trim) the unique patented Coopers' system	Reinforced Suspended ceiling with recessed shadow gap



The engineered design of the curved glazed façade required a bespoke fire barrier to provide boundary protection.

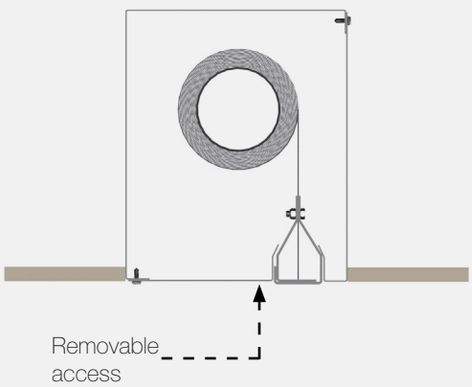
The FireMaster® fire curtains installed in this example illustrate how the headboxes and side channels are invisibly concealed and the entire system installed cannot be seen until deployed.



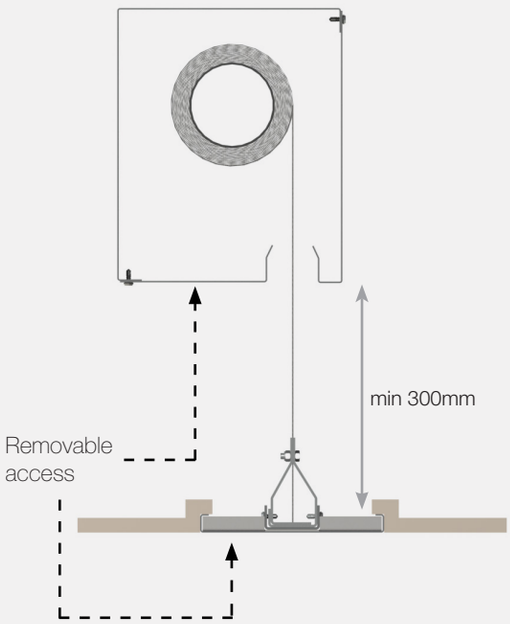
HOW IS THE BARRIER ACCESSED?

(Remember, regular service and maintenance is a legal mandatory requirement)

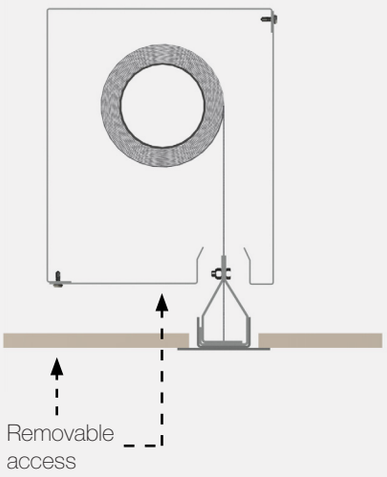
Flush fit



Suspended ceiling with SLAT™ (Self Levelling Access Trim) the unique patented Coopers' system



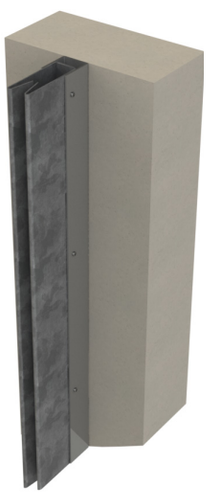
Reinforced Suspended ceiling with recessed shadow gap



SIDE CHANNELS

Side channels seal the curtain fabric and the building. Fire can cause extreme positive or negative pressures. Side guide channels are made from galvanised steel (available in standard RAL colours) that can be either surface mounted or recessed into the wall structure to give a flush finish. Coopers' unique fabric locking system has been developed and proven over many years.

End Fixed



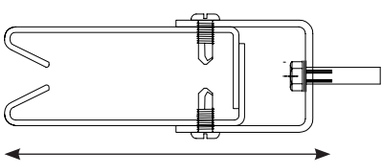
Face Fixed



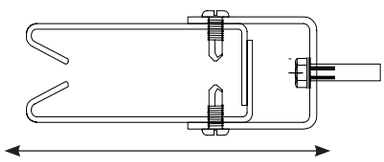
Recessed Fixed
(Recommended 5mm gap between side guide and building material)



STANDARD SIDE GUIDE



OPTIONAL FOR CURTAINS WITH LESS THAN 4M DROP

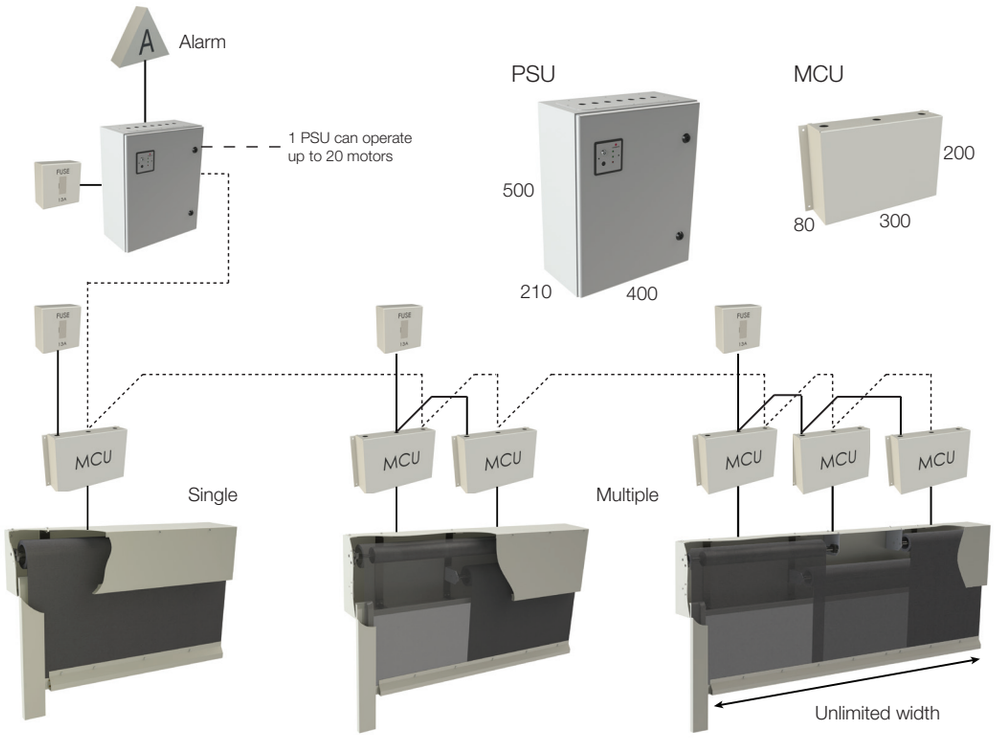


CONTROL OPTIONS

Controls are custom made to be fully compatible with both existing alarm systems and the number of barrier units installed, whether single or multiple roller assemblies.

Coopers also design and build Grouped Control Panels to accomodate customers requirements. This allows all controls to be installed within a single enclosure for easier installation, commissioning, servicing and reduction in field wiring. Please contact Coopers for more information on Grouped Controls.

When an alarm signal is detected, the control panel will automatically trigger all the Barrier systems to deploy by controlled descent using our unique Total Gravity Fail-Safe (TGFS) system.



PRODUCT SPECIFICATION

The complete system is assessed for 240 minutes (4 hrs) and 120 minutes (2hrs) in accordance with BS 476-22.8:1987, 120 minutes (2hrs) fire resistance and to the latest BS EN 1634-1:2008 and is classified as E120 EW30 S_a C2 Grade 3 in accordance BS EN 13501 2:2007+A1:2009. It also meets the requirement of BSI PAS 121:2007 (where applicable).

The fabric is additionally tested for fire propagation to BS 476 6:1989+A1:2009 and surface spread of flame to BS 476-7:1997 to achieve National Class '0' in accordance with A13(b) of Approved Document B (Volumes 1 & 2) 2006 Edition 'Fire Safety' to England & Wales Building Regulations 2000.

EFP™ 4/1000 is a long established, high performance fire textile composite material of woven glass fibre with a high performance coating, containing a micronised aluminium enriched polymer, which when exposed to fire, is absorbed into the surface of the curtain to provide high temperature performance. This coating was developed and used first by Coopers Fire.

DON'T BE LIABLE

Fire Curtain Barriers MUST have a 'Total' Fail-Safe by Gravity* in the event of total mains, wiring, short circuit and/or system corruption.

Coopers Fire, with their unique patented Total Gravity Fail-Safe (TGFS) system, are the only supplier with Independant Third Party Certification by a UKAS approved certification body to BSI PAS 121:2007 and now meets essential requirements of BS 8524-1:2013.

CONTROLS SPECIFICATION

POWER SUPPLY
230V AC 50Hz dedicated supply via 13Amp un-switched fused spur

BATTERIES
2 x 12V 12A/h lead acid batteries

ALARM SIGNAL
Normally closed volt-free contacts. Open on activation

TEST FACILITY
Zone Control Panel (ZCP) located on front of Power Supply Unit (PSU)

DISPLAY
Power ON, Alarm ACTIVE and Alarm OFF status LED's on ZCP

POWER SUPPLY SIZE
400mm(w) x 500mm(h) x 210mm(d)

MOTOR CONTROL UNIT SIZE
300mm(w) x 200mm(h) x 80mm(d)

FM 22356 LQR 4007272

CPD ACCREDITATIONS



For information, to get a quote
or to book a CPD workshop

Please call:
+44 (0)2392 454 405

or E-mail: info@coopersfire.com

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Coopers Fire Ltd has a policy of continuous product improvement.

As such we reserve the right to change design and specifications without prior notice.

Please check our website for the latest information.

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